

On the road to interactive LLM-based systematic mapping studies

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ABSTRACT

Context: The research volume is continuously increasing. Manual analysis of large topic scopes and continuously updating literature studies with the newest research results is effort intensive and, therefore, difficult to achieve.

Objective: To discuss possibilities and next steps for using LLMs (e.g., GPT-4) in the mapping study process.

Method: The research can be classified as a solution proposal. The solution was iteratively designed and discussed among the authors based on their experience with LLMs and literature reviews.

Results: We propose strategies for the mapping process, outlining the use of agents and prompting strategies for each step.

Conclusion: Given the potential of LLMs in literature studies, we should work on a holistic solutions for LLM-supported mapping studies.

1. Introduction

The mapping study methodology is frequently used in the context of software engineering. Two guidelines for mapping studies (MSs) were proposed and are, to date, cited more than 7000 times.¹ The purpose of mapping studies is to classify research and observe trends. In contrast, systematic literature reviews (SLRs) focus on synthesizing evidence. Given the purpose of mapping studies, a more extensive set of studies should be targeted, as otherwise, it would make sense to conduct an SLR directly.

This article discusses the possibilities of using LLMs throughout the mapping process, proposing strategies for using LLMs in each research step of MSs. There are various motivations for using LLMs in MSs: (1) An increased number of published papers to be analyzed; (2) Conducting mapping studies on a larger scope; (3) Getting additional research design ideas by interacting with the LLM; (4) Reduced effort allows updating mapping studies more regularly. The solution described here is a direction for researchers in the software community to work on solutions as a community, as a reliable and intensively evaluated solution will likely result from a community effort. This article may serve as a starting point for discussing using LLMs in MSs, working towards a holistic solution, and evaluating the solution in the SE context.

We describe the solution and provide references to relevant studies. Our solution is inspired by [1], where the authors propose a system where testers interact with a search-based software test system,

observing the output and the tester providing feedback to the search-based system based on candidate solutions. In our solution proposal, we follow this human-in-the-loop approach outlined by [1]. To assess and react appropriately to the output of the LLM-system, it is essential that reviewers (a) are well educated in using the mapping study method, and (b) be experts in the topic they are reviewing.

Khalil et al. [2] reviewed machine learning tools for literature studies. The tools were assessed prior to ChatGPT's release. Therefore, the tools required manual labeling (e.g. during paper screening), whereas a LLM like ChatGPT can produce a descriptive assessment outcome including rationales and text fragments. Similarly, the framework by [3] focusing on paper screening needed to be trained based on the users' labels.

2. LLM-supported mapping process

This section discusses how to incorporate LLMs in every step of the mapping process, which is presented in [4]. Fig. 1 presents each step of the review process and briefly outlines the input and actions done by the user and the output of the LLM. For all the steps outlined below we would like to highlight that the researcher should have an initial idea of possible search terms, inclusion criteria or extraction items in order to validate the outputs. Specifically, research problems and goals should be driven by relevant practical needs.

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¹ According to Google Scholar, August 2024.

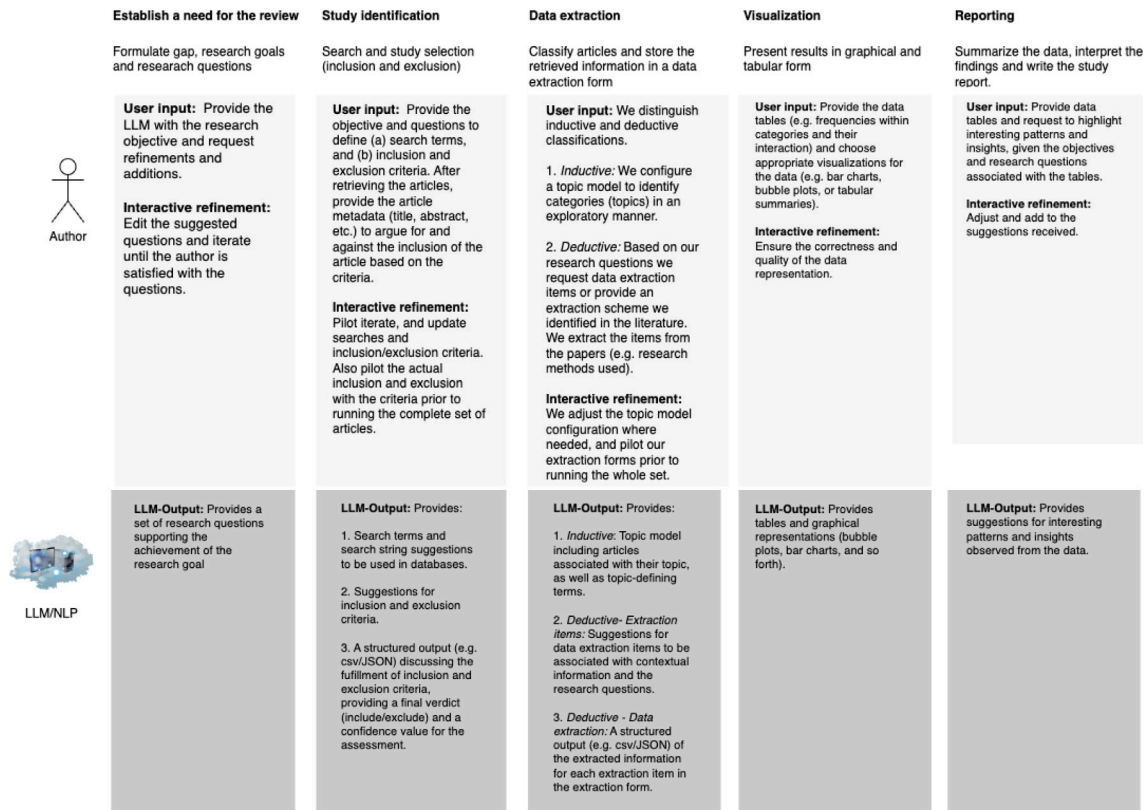


Fig. 1. The mapping process with LLM support.

2.1. Establishing a need for the map

At this stage, we provide the LLM with our research objectives and contextual information (e.g., paper abstract) to elicit proposals for research questions and complementary information (e.g., additional objectives). We edit the questions as input for the next stage.

2.2. Study identification

2.2.1. Search

Proposed strategy: Search aims to identify relevant documents concerning the research objective. Along with defining research objectives, the search is a complex and iterative process. Creating a search strategy from a search intent involves understanding the searcher's underlying intentions. The strategy must be both transparent and reproducible. Even though semantic searches² are becoming more common, the need for reproducibility supports using Boolean searches. Boolean searches necessitate formulating a search strategy using Boolean expressions and operators. The prerequisite is the identification and combination of relevant terms. We advocate for an autonomous agent with Human-in-the-Loop to meet the demand for reproducibility, account for human involvement in the process, and leverage the potential of new technological developments. We propose an architecture that places human intervention at the center and considers three agents:

1. **Keyword Identification Agent:** The Keyword Identification Agent identifies relevant terms and provides them for further research. Language models can support the search for semantically similar terms. Semantic similarities can, for instance, help find terms commonly used in the past but have largely been replaced by other terms today. This step also serves to solidify the focus of

the study. For example, in 3D printing, the searcher's intention may be to search at the conceptual term level, the level of subtypes (e.g., Stereolithography), or supertypes (e.g., Rapid Prototyping).

2. **Semantic Search Agent:** The Semantic Search Agent uses a Retrieval-Augmented Generation (RAG) architecture to suggest relevant documents based on semantic similarities. In this context, it may be helpful to consider a graph database as a storage component, allowing for the incorporation of links between documents (e.g., through citations). This step does not select documents but adjusts the search strategy. Relevant search terms are then extracted again from the selected documents.
3. **Search Strategy Agent:** The Search Strategy Agent formulates the search strategy used to conduct the research.

The Keyword Identification Agent and the Semantic Search Agent thus specifically support a citation pearl growing strategy.

Relevant literature: Wang et al. [5] created prompts of different complexities, the most complex including a search example and requesting the use of PICO. Also, prompts were refined to request modifications to increase the relevance of elements. In addition, so-called guided prompts were formulated that separate the search construction process into sub-steps. The authors used existing SLRs as a basis to assess the generated searches. The results show the potential to use ChatGPT for search generation. The F-Measure shows better results for author-based queries; however, GPT-generated queries result in less recall, i.e. more relevant papers are missed. Additional interesting findings were: (1) The use of PICO harms recall; (2) the use of existing SLRs as a basis to assess the generated searches; (3) requesting refinements reduces recall and improves precision.

2.2.2. Inclusion and exclusion

Proposed strategy: Defining inclusion and exclusion criteria is crucial for capturing the researcher's intention. Traditional prompt engineering techniques may reach their limits here. We argue that search

² Underlined words are defined in the supplementary material.

intentions are often initially latent and manifest during the search process. Therefore, a continual learning system can be superior to hard-coded prompts. Users' preferences can be derived from document inclusion and exclusion. Frameworks like DSpy can also be helpful by optimizing prompts. Learning approaches require larger datasets, which researchers need to create iteratively.

From a technical perspective, including and excluding documents is a classification problem. Researchers classify documents as relevant or not relevant. However, this approach lacks transparency, traceability, and human oversight. Therefore, language models have to explain the reasons for inclusion and exclusion. Chain-of-thoughts prompting is a possible approach to help researchers understand the decision-making process of the LLM. From our perspective, citations are indispensable. They allow the verification of arguments and increase traceability.

Relevant literature: Huotala et al. [6] used GPT to simply abstracts of papers and compare the paper selection done by GPT with human raters. As found by the authors, a similar performance between humans and the LLMs is achieved when using One-shot, Few-shot, or Few-shot with Chain-of-Thought prompting. The authors also found improved performances when using GPT-4 compared to GPT-3.5 when using prompt optimization. Zero-shot prompting was not performing well compared to other prompt optimization approaches. The authors also highlight the benefit of having seed papers or prior SLRs as input for prompting. Guo et al. [7] repeated the selection process of existing SLRs, highlighting that GPT was good at excluding irrelevant papers but not as good at achieving a high recall. They highlighted the value of GPT as a support tool for reviewers to save time.

2.3. Data extraction and classification

Before discussing the data extraction and classification strategies, we would like to highlight that, given the degree of automation used, it is now possible to go beyond adaptive reading depth during manual screening and consider the complete papers as input for the analysis for deductive coding.

1. **Inductive coding:** In inductive coding, the classification emerges as a consequence of the coding process. We suggest using topic modeling with the titles and abstracts as input to identify the topics and clusters of papers. A typical process is to create embeddings, reduce dimensions, cluster embeddings, and create topic representations. Such a process is, for example, supported by Bertopic, which supports a modular framework, allowing the exchange of the approaches used, for example, for clustering.
2. **Deductive coding:** Given is a data extraction scheme (e.g., SWE-BOK categories). As was learned in the context of screening, examples are important. Hence, we suggest using One-shot or Few-shot prompting. As the complete PDF document is analyzed, a RAG architecture may be helpful, as it first identifies the most relevant parts of a document after document splitting before prompting the OpanAI API.

Relevant literature: A solution for inductive coding was proposed by Wang et al. [8]. They used the aforementioned Bertopic library to review interdisciplinary research. They compute topic diversity and cohesion of topics to learn about interdisciplinary integration of publications. Besides showing hierarchical clusters of topics, feature words, and c-TF-IDF scores for terms within topics, they also included a temporal analysis. Petersen [9] used GPT-4 to judge whether studies are indeed case studies as claimed by study authors. The result showed that GPT-4 outperformed the authors (both primary and secondary studies).

2.4. Visualization

ChatGPT already provides options for data visualization by generating source code and visualizing data. However, it is noticeable that further specialized applications are being developed. An example of this is LIDA from Microsoft.

For exploratory analysis Bertopic may be used, which allows for the visualization and exploration of document landscapes.

2.5. Reporting

To support the reporting of studies, we suggest providing GPT with tabular results and result visualizations and request to highlight exciting patterns and observations in the data (e.g., with a specific focus on spotting research gaps).

3. Reflections

Study validity: Former studies have shown that GPT achieved similar or even better performance than humans. Publication bias and limited studies currently restrict our understanding of LLMs' reliability in literature reviews. The rapid evolution of LLMs, exemplified by Claude.ai and the upcoming GPT-o1, may quickly render current evaluations obsolete. This underscores the need for a conceptual framework to guide future research and applications of LLMs in software engineering (SE). Many existing studies are from outside SE, highlighting the importance of developing SE-specific solutions. We aim to inspire further research on LLMs' performance in SE literature studies, acknowledging that findings from current models may not apply to future iterations.

Complementary Tools: Tracing tools such as LangSmith allow for tracing (debugging, testing) individual steps in LLM-based systems, which is of use when engineering a systematic mapping tool. WebVoyager [10] can search the entire internet using a visual approach. As WebVoyager is not limited to research references, it may be of interest for grey literature reviews.

We suggest two research directions:

- Improving individual steps and evaluating different strategies (e.g., prompting) to optimize them.
- Build a prototype representing the overall mapping process to gather further ideas for supporting mapping studies end-to-end.

CRedit authorship contribution statement

Kai Petersen: Writing – review & editing, Writing – original draft, Conceptualization. **Jan M. Gerken:** Writing – review & editing, Writing – original draft, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.infsof.2024.107611>.

Data availability

No data was used for the research described in the article.

References

- [1] Bogdan Marculescu, Simon Poulding, Robert Feldt, Kai Petersen, Richard Torkar, Tester interactivity makes a difference in search-based software testing: A controlled experiment, *Inf. Softw. Technol.* 78 (2016) 66–82.
- [2] Hanan Khalil, Daniel Ameen, Armita Zarnegar, Tools to support the automation of systematic reviews: a scoping review, *J. Clin. Epidemiol.* 144 (2022) 22–42.
- [3] Van De Schoot, et al., An open source machine learning framework for efficient and transparent systematic reviews, *Nat. Mach. Intell.* 3 (2) (2021) 125–133.
- [4] Kai Petersen, Sairam Vakkalanka, Ludwik Kuzniarz, Guidelines for conducting systematic mapping studies in software engineering: An update, *Inf. Softw. Technol.* 64 (2015) 1–18.
- [5] Shuai Wang, Harris Scells, Bevan Koopman, Guido Zuccon, Can ChatGPT write a good boolean query for systematic review literature search? in: *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2023, pp. 1426–1436.
- [6] Aleksi Huotala, Miikka Kuuttila, Paul Ralph, Mika Mäntylä, The promise and challenges of using LLMs to accelerate the screening process of systematic reviews, 2024, arXiv preprint [arXiv:2404.15667](https://arxiv.org/abs/2404.15667).
- [7] Eddie Guo, Mehul Gupta, Jiawen Deng, Ye-Jean Park, Mike Paget, Christopher Naugler, Automated paper screening for clinical reviews using large language models, 2023, arXiv preprint [arXiv:2305.00844](https://arxiv.org/abs/2305.00844).
- [8] Zhongyi Wang, Jing Chen, Jiangping Chen, Haihua Chen, Identifying interdisciplinary topics and their evolution based on BERTopic, *Scientometrics* (2023) 1–26.
- [9] Kai Petersen, Case study identification with GPT-4 and implications for mapping studies, *Inf. Softw. Technol.* 171 (2024) 107452.
- [10] Hongliang He, Wenlin Yao, Kaixin Ma, Wenhao Yu, Yong Dai, Hongming Zhang, Zhenzhong Lan, Dong Yu, WebVoyager: Building an end-to-end web agent with large multimodal models, 2024.